

Capstone Project Plan

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Introduction and Background

Chronic diseases such as diabetes, obesity, cardiovascular disease, and cancer remain leading causes of death and disability in the US. This project utilizes data from the *Centers for Disease Control and Prevention's (CDC) Chronic Disease Indicators (CDI)*, a standardized dataset developed in collaboration with state epidemiologists and public health officials.

The reason to use this dataset is mainly because of the multiple authoritative sources, including BRFSS, NVSS, ACS, ensuring consistency and reliability across measurements. Moreover, we aim to assess risk prediction, evaluate the feasibility of cross-disease comparisons, and identify disparities across demographic subgroups based on the exploration and analysis of this dataset.

Dataset Description

There are over 300,000 observations and 32 features included in this dataset across all U.S. states and territories. The dataset spans multiple years, from 2015 to 2022, enabling longitudinal analysis of chronic disease trends over time and the assessment of long-term public health interventions.

Besides that, each observation corresponds to a unique combination of health indicators (Question), state (Location), demographic group (StratificationCategory), help to report indicators such as prevalence rates, hospitalizations, and behavioral risk factors stratified by sex, age, and race/ethnicity.

Questions and Problems

- Problems

The chronic disease indicators dataset, while rich in breadth and depth, presents several analytical challenges that must be addressed to ensure valid, interpretable, and actionable results.

First, one of the major issues lies in unit and type inconsistencies across indicators. The target variable: `DataValue` with a large range, which means the health metrics in the dataset are reported in different units—including percentages, crude rates, age-adjusted rates, and case counts per population—which are not directly comparable. This heterogeneity limits the ability to conduct meaningful cross-disease comparisons or aggregated scoring without standardization techniques.

Second, the dataset draws from multiple sources, each with different sampling strategies, coverage, and target populations. This variability introduces potential systematic bias or noise into the analysis. Indicators from different sources, even when measuring the same health outcome, may not be directly comparable unless properly accounted for during analysis.

A notable challenge in this study arises from the substantial proportion of missing values in the primary outcome variable, `DataValue`, which represents the observed measurement for each chronic disease indicator. Specifically, approximately 32.35% of records lack valid `DataValue` entries, significantly affecting the overall completeness of the dataset. The absence of these values prevents meaningful calculation of means, comparisons across groups, and construction of predictive models unless properly handled.

- Questions

In light of the above challenges, this study aims to explore a series of structured research questions to guide analysis and ensure actionable outcomes.

The first focus is on Invasive Cancer (All Sites Combined): Is there a statistically significant difference in the incidence and mortality rates of invasive cancer across different genders, U.S. states, and years?

Next, the study investigates lung cancer deaths and examines whether disparities exist across population subgroups and regions, as well as whether there are trends or shifts over time. Question: Does the mortality rate of lung and bronchial cancer vary significantly by gender, location, and over time in the United States?

The third research direction concerns temporal trends. Has the prevalence of major chronic diseases increased or declined over the past decade? Finally, the project explores predictive risk classification. Can we accurately predict whether a particular state and demographic group will fall above the national median? If successful, this modeling approach could inform targeted surveillance and policy intervention strategies.

Methods

- Chi-Square Test of Independence (Sex vs Topic, Sex vs State, and Topic vs State). Log-linear modeling using a three-way contingency table ($\text{Sex} \times \text{Topic} \times \text{State}$)
- ANOVA: A good method to identify the indicators causing the variance of data. Determine whether the mean of target value significantly differs across various groups such as geographic regions, periods, or demographic categories.
- Logistics Regression: Build a classification model to predict whether a specific region or demographic group is in a high-risk category (higher than national median value)

References

Metadata, U.S. Chronic Disease Indicators, 2025. Retrieved from
<https://catalog.data.gov/dataset/u-s-chronic-disease-indicators>